
Assessment CompTia A+

A+ - Troubleshooting And Helpdesk

Section A:- Concept Application

SCENARIO-1

You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.



Process for Selecting Compatible Hardware:-

1. **CPU:** Check the CPU's socket type, such as AM5 or LGA1700, and select a motherboard with the same socket.
2. **Motherboard:** Check that it supports the CPU, required RAM type, storage interface, and has enough ports and expansion slots.
3. **RAM:** Check the motherboard's supported RAM type (such as DDR5), maximum capacity, and supported speed. For an office PC, 16 GB RAM is usually a practical choice.
4. **Storage:** Select an SSD that matches the motherboard's supported interface, such as M.2 NVMe or SATA. An NVMe SSD is a good choice for faster boot and application loading.
5. **Check Compatibility:** Use the motherboard manufacturer's CPU support list, RAM compatibility/QVL, and storage specifications. Also check BIOS requirements for the CPU.
6. **Final Confirmation:** Make sure the CPU socket, RAM generation, and storage connector all match the motherboard. After assembly, enter BIOS/UEFI to confirm that the CPU, RAM, and storage are detected correctly.

SCENARIO-2

You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.



HDD: HDDs use spinning magnetic disks to store data. They are cheaper and provide large storage capacity, but they are slower and have moving parts, which can make them less suitable for fast office work.

SSD: SSDs use flash memory and have no moving parts. They are much faster and more reliable than HDDs, making applications, Windows, and file access quicker.

NVMe: NVMe SSDs are a faster type of SSD that connects through the PCIe interface. They offer higher read/write speeds and lower latency than SATA SSDs.

Recommendation: For office and productivity work, I would recommend an **NVMe SSD**. It provides fast boot times, quick application loading, and responsive multitasking. Although it may cost slightly more than an HDD, its speed and reliability make it suitable for modern office workstations.

SCENARIO-3

You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.



GPU:

The GPU handles graphics processing and helps design software render images, 3D models, and visual effects smoothly. A suitable GPU provides better performance and reduces the workload on the CPU during graphics-intensive tasks.

Cooling System:

The cooling system removes heat produced by the GPU and CPU during long working hours. Fans and heatsinks keep temperatures under control and prevent overheating, thermal throttling, and sudden performance drops.

Conclusion:

A good GPU with proper cooling helps the workstation maintain stable and consistent performance during several hours of continuous design work.

SCENARIO-4

You are assembling a PC from components sourced from three different vendors.

Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.



Hardware compatibility is important because all computer components must work together properly. For example, the CPU must match the motherboard socket, RAM must be supported by the motherboard, and the storage device must use a compatible connection such as SATA or M.2.

Incompatible components, can cause problems such as the computer not starting, RAM not being detected, storage devices not working, overheating, or poor performance. Physical size and power requirements can also cause problems if components do not fit the case or the power supply is insufficient.

I would, check the motherboard manufacturer's compatibility list, CPU socket, RAM type, storage interface, power requirements, and physical dimensions before buying or assembling the parts.

SCENARIO-5

You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.



1. **Check power needs:** Add the power requirements of the CPU, GPU, storage, and other parts.
2. **Choose PSU wattage:** Select a PSU with some extra capacity, usually around 20–30% headroom.
3. **Check PSU quality:** Use a reliable PSU with proper efficiency certification and safety protections.
4. **Connect motherboard:** Plug in the 24-pin ATX and 4/8-pin CPU power connectors.
5. **Connect GPU:** Connect the required 6/8-pin PCIe power cable if the GPU needs one.
6. **Connect storage:** Connect SATA power to SATA HDDs/SSDs.
7. **Safety check:** Make sure cables are firmly connected and there are no loose wires or damaged connectors.

Final check: Keep the PSU switch OFF while checking everything, then switch it on and perform the first boot.

SCENARIO-6

You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.



1. **Power on the PC:** Press the power button and check that fans start and the display appears.
2. **Check POST:** The system should complete the startup check and show the manufacturer logo or BIOS screen without errors.
3. **Check beep/LED codes:** Listen for beep sounds or check motherboard debug LEDs for any warning.
4. **Check components:** In BIOS, confirm the CPU, RAM, and storage drive are detected correctly.
5. **If POST fails:** Turn off the PC and check power cables, RAM seating, GPU connection, and other components.
6. **Test step by step:** Remove unnecessary devices and reseat or test components one at a time to find the faulty part.

Section B:- Practical Coding Task

TASK-1

Component Identification & Functions



Task 1: Component Identification & Functions

Identify and document the functions of the core hardware components in a desktop system.
(Foundational — direct identification of a single concept.)
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Power Supply (SMPS)

CPU (with Cooler)

Motherboard

GPU (Graphics Card)

SSD (Primary Storage)

RAM

HDD (Secondary Storage)

Components at a Glance

- SMPS (Power Supply)
- CPU (with Cooler)
- Motherboard
- RAM
- GPU (Graphics Card)
- SSD (Primary Storage)
- HDD (Secondary Storage)
- Cooling System (CPU Cooler)

Component	Image / Identification	Make / Model or Key Specification	Function
Motherboard		ASUS Prime B660M-A	Connects all hardware components and allows them to communicate with each other.
CPU		Intel Core i5-12400 (6 cores, 12 threads)	Processes instructions and performs calculations. It is the main processing unit of the computer.
RAM		16 GB DDR4, 3200 MHz (Corsair Vengeance LPX)	Temporarily stores data and programs that the CPU is currently using.
GPU		NVIDIA GeForce RTX 3060 (12 GB)	Processes graphics, videos and gaming visuals.
Storage Device – SSD		Samsung 870 EVO 500 GB (SATA)	Stores the operating system, applications and files. Provides faster data access than an HDD.
Storage Device – HDD		Seagate Barracuda 1 TB (7200 RPM)	Provides large-capacity storage for files, videos and backups.
Cooling System		Intel stock CPU cooler (air cooling)	Removes heat from the CPU and helps prevent overheating.

Conclusion: These components work together to make the desktop computer operate. The motherboard connects them, the CPU processes data, RAM provides temporary working space, the GPU handles graphics, storage keeps data permanently, and the cooling system controls heat.

- Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.
- Note the make/model or key specification of each component where visible.
- Present the findings as a labeled component list or table

Component	Function	Make/Model or Key Specification
Motherboard	Connects all hardware components and allows them to communicate with each other.	ATX/Micro-ATX; model visible on board
CPU (Processor)	Performs calculations and executes instructions. It is the main processing unit of the computer.	Intel Core i5 / AMD Ryzen 5, etc.
RAM	Temporarily stores data and programs currently being used. More RAM helps with multitasking.	8 GB / 16 GB DDR4 or DDR5
GPU (Graphics Card)	Processes images, videos, and graphics for the monitor. Useful for gaming and graphic applications.	NVIDIA GeForce / AMD Radeon; model as visible
Storage Device	Permanently stores the operating system, software, and personal files.	512 GB SSD / 1 TB HDD, etc.
Cooling System	Removes heat from the CPU and other components to prevent overheating.	CPU fan/heatsink or liquid cooler

TASK-2

➤ Motherboard Ports & Socket Identification

- Identify and label all major motherboard ports (SATA, power, USB headers, etc.).

Port / Connector	Function
24-pin ATX Power Connector	Supplies main power from the SMPS to the motherboard.
4/8-pin CPU Power Connector	Provides power to the CPU.
SATA Ports	Connect SATA HDDs, SSDs, and optical drives.
M.2 Slot	Connects M.2 SSDs, including NVMe SSDs.
RAM/DIMM Slots	Hold the RAM modules.
PCIe x16 Slot	Used mainly for a graphics card (GPU).
PCIe x1 Slots	Used for expansion cards such as Wi-Fi or sound cards.
USB Headers	Connect front-panel USB ports of the computer case.
Front Panel Header	Connects the power switch, reset switch, power LED, and HDD LED.
CPU Fan Header	Connects the CPU cooling fan.
System Fan Header	Connects additional case fans.
Front Audio Header	Connects the case's front headphone and microphone ports.
CMOS Battery	Maintains BIOS/UEFI settings and system clock when the PC is off.
Rear USB Ports	Connect external USB devices such as keyboards, mice, and storage drives.
Ethernet/LAN Port	Connects the computer to a wired network or Internet.
Audio Ports	Connect speakers, headphones, and microphones.
HDMI/DisplayPort	Provides display output when supported by the CPU/motherboard.

- Identify all expansion slots (PCIe, M.2, etc.) and their purpose.

Expansion Slot	Purpose
PCIe x16	Used mainly for graphics cards (GPU).
PCIe x1	Used for Wi-Fi, sound, network, and other small expansion cards.
PCIe x4 / x8	Used for high-speed expansion cards such as storage or network cards.
M.2 Slot	Used for M.2 SSDs, especially fast NVMe SSDs.
DIMM Slots	Used to install RAM modules.

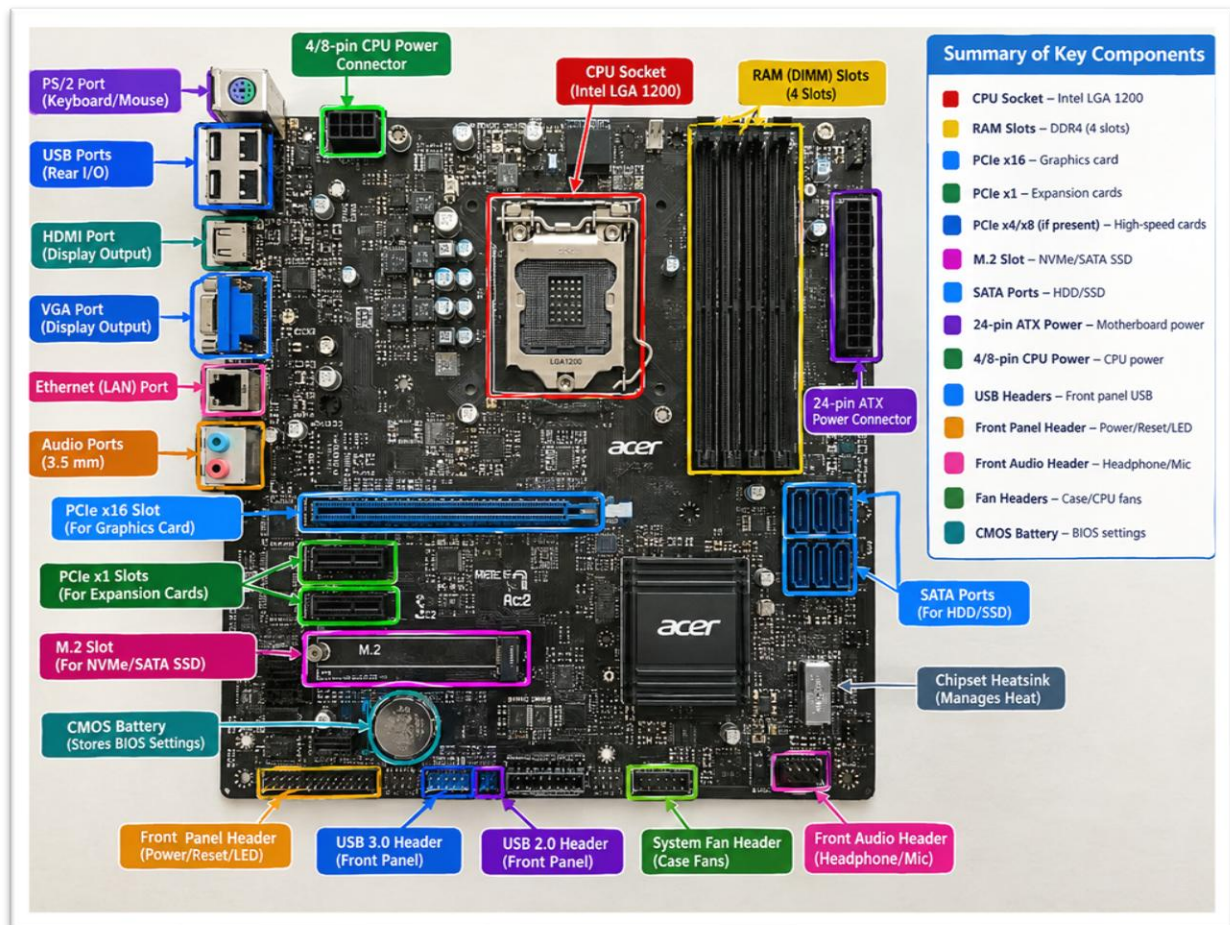
- Identify the CPU socket type and note which CPU families are compatible with it.

For the **ACER DESKTOP-5BA7NK7** :

CPU Socket Type: Need to check the exact Acer model/motherboard

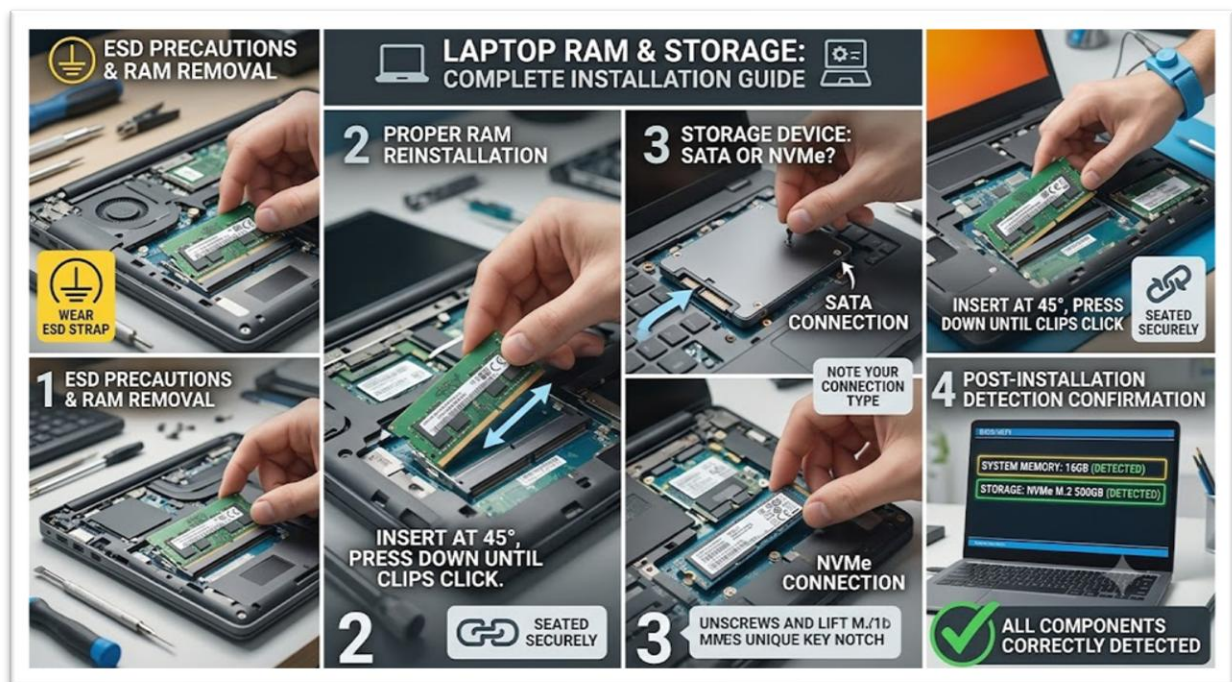
Compatible CPU Families: Depends on the socket and motherboard chipset.

- Photograph or diagram the motherboard with labels



TASK-3

RAM & Storage Installation



1. Remove RAM

Shut down the PC and unplug the power cable. Wear an **anti-static wrist strap** and touch the metal case to discharge static electricity. Open the RAM clips and carefully remove the RAM module.

2. Reinstall RAM

Align the notch on the RAM with the slot. Press the RAM down firmly until both clips lock automatically. Check that the RAM is fully seated.

3. Remove and Reinstall Storage

Identify the storage type: SATA SSD/HDD or M.2/NVMe SSD. For SATA, disconnect the SATA data and power cables. For NVMe, remove the M.2 screw, take out the SSD, reinstall it, and tighten the screw.

4. Check Components

Close the case and reconnect the power. Start the PC and check BIOS/UEFI or Windows Task Manager/Device Manager. Confirm that the installed RAM and storage device are detected correctly.

Result: RAM was properly seated and the storage device was reinstalled and detected successfully.

TASK-4

Full Desktop PC Assembly

- Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.
- Follow correct installation sequence and ESD/safety precautions throughout.
- Document each step with labeled photos or diagrams.
- Power on the system and confirm it passes POST successfully



1. **Install components** – Install the CPU, cooler, RAM, and storage on the motherboard.
2. **Install inside cabinet** – Secure the motherboard, PSU, and GPU, then connect all required cables.
3. **Follow safety** – Use ESD protection, keep power disconnected during assembly, and handle components carefully.
4. **Test and document** – Power on the PC, confirm successful **POST/BIOS**, and take labeled photos of each step.

Section C:- Mini Capstone Project

Mini Project: Workstation Deployment & Boot-Failure Diagnosis

Objective:

ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.

- **Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.**

1. Three hardware component to check first are:

1. **Power Supply Unit (PSU):** First, check the PSU and make sure it is switched on. Check that the 24-pin motherboard power cable and CPU power cable are connected properly. A loose or missing power connection can prevent the PC from starting.
2. **RAM:** Check whether the RAM module is installed correctly and firmly in the correct slot. Loose, damaged, or incompatible RAM can cause no display or POST errors. Removing and reinstalling the RAM can help identify the problem.
3. **CPU:** Check that the CPU is installed correctly and that the CPU power connector is connected. Also check that the CPU cooler is properly fitted. Incorrect CPU installation or a missing CPU power connection can stop the computer from booting.

- **Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.**

2. Motherboard-CPU Compatibility

Motherboard–CPU compatibility is important because the CPU must match the motherboard’s socket and chipset.

In the system I assembled, I checked that the CPU was supported by the motherboard before installation. The compatible CPU allowed the system to recognize the processor and communicate properly with the RAM, storage, and other components. If the CPU were incompatible, the PC might not boot or pass POST successfully.

- **Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.**

3.Verifying RAM and storage Installation

I would check that the RAM is firmly seated and the storage device is properly connected. After powering on the computer, I would check the BIOS/UEFI to confirm the correct RAM capacity and storage drive are detected. A successful POST without error messages confirms that the components are installed correctly.

- **Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.**

4. Hardware Installation Checklist

1. Check compatibility – Confirm that the CPU, RAM, storage, GPU, and motherboard are compatible before installation.
 2. Check connections – Make sure RAM, storage, GPU, and power cables are properly seated and connected.
 3. Follow ESD safety – Use an anti-static wrist strap or touch a grounded metal surface before handling components.
 4. Test before closing – Power on the PC and check BIOS/POST to confirm all hardware is detected correctly.
 5. Document the installation – Record the installed components and any issues for future maintenance.
- **Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.**

5. Why SSD is better than HDD for the office lab

SSDs are preferred over HDDs in modern office workstations because they provide faster boot times, quicker application loading, and faster file access. SSDs also have no moving parts, making them more resistant to physical shock and generally more reliable than mechanical HDDs.

For the ABC Solutions computer lab, SSDs would help staff and students start computers quickly, open office applications faster, and reduce waiting time during daily work. This can improve productivity and reduce downtime when multiple workstations are being used.

Section D:- AI-Augmented Learning

STEP 1 · BUILD WITH AI

- Guides a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly, in the correct priority order.
- Covers at minimum the motherboard, RAM, storage, and PSU as possible failure points.
- Is presented as a clear checklist or simple decision flow a technician could follow on-site.
- Is applied to the ABC Solutions scenario from Section C to show which step would have caught the reported issue.

1. Exact Prompt Given to AI

“Create a step-by-step troubleshooting checklist for a junior technician to diagnose a desktop PC that fails to boot immediately after assembly. Put the steps in the correct priority order. Include the motherboard, RAM, storage device, and PSU as possible failure points. Make it a simple checklist that can be followed on-site. Apply the checklist to the ABC Solutions scenario and explain which step could identify the reported issue.”

2. AI's Original Checklist

1. Check PSU and power cables.
2. Check motherboard connections and LEDs.
3. Reseat and test RAM.
4. Check CPU installation and CPU fan.
5. Check storage device and BIOS detection.
6. Check GPU and display connection.
7. Test POST and identify error codes.

STEP2 -TEST AND DEBUG (Without AI)

1. **Safety first:** Switch off and unplug the PC. Use ESD protection before touching internal components.
2. **Check PSU:** Verify the PSU switch, power cable, 24-pin motherboard cable, and CPU power cable.
3. **Check motherboard:** Check mounting, connections, diagnostic LEDs, and visible damage.
4. **Check RAM:** Reseat RAM and test one module at a time if necessary.
5. **Check CPU:** Verify CPU seating, cooler installation, and CPU_FAN connection.
6. **Check storage:** Reseat SATA/NVMe storage and check whether it is detected in BIOS.
7. **Test POST:** Power on and check for successful POST, display, beep codes, or diagnostic LEDs.

3. What I Changed:

- The AI checklist did not clearly include ESD safety before handling internal components.
I added ESD protection and power disconnection as the first step to prevent damage or electrical hazards.
I also kept physical hardware checks before BIOS/software checks because the PC had just been assembled.
This makes the troubleshooting process safer and more suitable for a junior technician.